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MILITARY SPECIFICATION

WELDING SET, ARC, GAS SHIELDED, CONSUMABLE ELECTRODE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers arc welding sets using gas shielded, consumable electrodes in conjunction with direct current power supplies for welding ferrous and nonferrous metals in the gas metal arc welding (GMAW) processes with equipment and accessories specified herein.

1.2 Classification. The sets shall be of the following types, classes, and styles. The type, class, and style to be furnished shall be as specified (see 6.2.1).

Type I - Push Wire Feed

Class A - Air-Cooled

Style 1 - Fine Wire
Style 2 - Heavy Wire
Style 3 - Flux Cored Wire
Style 4 - Compact Portable

Class B - Water-cooled

Style 1 - Fine Wire
Style 2 - Heavy Wire

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - Pull Wire Feed

Class A - Air-Cooled

Style 1 - Fine
Style 2 - Self-Contained Spool on Gun

Class B - Water-Cooled

Style 1 - Fine Wire

Type III - Push-Pull Wire Feed

Class A - Air-Cooled

Style 1 - Fine Wire
Style 2 - Air-Turbine Driven

1.2.1 Classification definitions. Classification by wire feed methods stipulated in this specification includes "push," "pull," and a combination of the two. The designation "push" is defined as the design whereby the electrode wire is pushed by the wire feeder through the wire conduit. The designation "pull" is defined as the design whereby the electrode wire is pulled by the wire feeder through the wire conduit. The following subparagraphs clarify and define each of three methods.

1.2.1.1 Push feed. Push feed types shall have a set of feed rolls which shall unreel and push the wire to the gun/torch. For purposes of identification in this specification, the gun/torch will be herein after referred to as "gun".

1.2.1.2 Pull feed. Pull feed types shall have a set of feed rolls in the gun itself which will pull the wire from the wire reel housing.

1.2.1.3 Push-pull feed. Push-pull types shall have two sets of synchronized, powered feed rolls. One set of powered feed rolls shall be located in the wire feed and control unit to unreel and push the wire through the cable to the gun. The gun shall have a set of powered feed rolls which shall be synchronized with those in the wire feed and control unit and shall deliver the welding wire to the gun barrel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-government documents which are current on the date of the solicitation.

AMERICAN WELDING SOCIETY, INCORPORATED (AWS)

A5.1 - Mild Steel Covered - Arc-Welding Electrodes.

(Application for copies should be addressed to the American Welding Society, Inc., ATTN: Sales Dept., 345 East 47th Street, New York, NY 10017.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B40.1 - Gauges, Pressure, Indicating Dial Type.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding sets shall be new and one of the manufacturer's current models capable of welding ferrous and nonferrous metals using the gas shielded, consumable electrode process in accordance with the requirements herein. The welding sets shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding sets. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1. Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International

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System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding set shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI Systems of Measurement.

3.2.2 Reclaimed materials. The welding set may contain reclaimed materials provided such materials will not jeopardize the welding set's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use in the welding set.

3.2.3 Energy efficiency. The welding set and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding set that present safety hazards. The safety devices shall not interfere with the operation of the welding set. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding set parts, components, mechanisms, and assemblies furnished on the welding set, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the welding set itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The welding set shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding set.

3.2.8 Environmental protection. The welding set shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the welding set shall not emit materials hazardous to the ecological system as prescribed by Federal statute in effect at the point of installation (see 6.4).

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3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The welding set shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding set to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective parts.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding set shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used in the welding set and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than .001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque to the drive motor without gear damage, failure, or premature deterioration and wear.

3.4 Components. The welding sets shall consist of, but not be limited to, the basic components listed below.

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a. Control unit.
b. Wire feeding device.
c. Gun.
d. Cable assembly.
e. Interconnecting cords, cables, and gas and water hoses.
f. Enclosure.

Other components shall be furnished as specified (see 3.4.7, 3.4.8, 3.4.9, 3.4.10, 3.4.11, and 3.4.12). When required, the welding set shall be compatible with a specific power supply as specified (see 6.2.1).

3.4.1 Control unit. The control of the wire feed and the welding current shall be electric, electronic, and electro-mechanical. The controls shall be designed to keep the wire feed constant regardless of normal friction imposed on the wire in the gun and cable assembly while welding. The controls shall be designed to withstand rough handling and abuse encountered in welding shops and outdoor construction. Wire connections shall be of the screw, spring clip, or soldered type, and all wires shall be numbered or color coded. The control shall be designed to operate in conjunction with a direct current (DC) constant voltage power supply. The control for the type II, class A, style 2 set shall be designed to operate in conjunction with either a constant voltage or a constant current power supply, and shall have the means to change from one to the other. When a switch is used for this purpose, it shall be located or protected to prevent accidental changing. The control for the wire feed unit of type III, class A, style 2 shall be by compressed air with an air regulator, air filter, and air lubricator located in a control cabinet. The input power for the control unit of the type I and II sets shall be 115-volt, alternating-current (AC), 60-hertz (Hz); except the type II, class A, style 2, may be either 115-volt, AC, 60-Hz, or DC or both. The voltage to the gun switch shall be reduced to 50 volts or less.

3.4.1.1 Wire feed speed control. The electric control shall have a wire feed speed control. The feed speed control shall be of the stepless mechanical or electric type. The feed speed control for the type II may be located either on the gun or the main control.

3.4.1.2 Solenoid gas valve. The electric control unit shall have a solenoid valve mounted in the control enclosure to start and stop the flow of the shielding gas. The type II and III sets may have a mechanical valve located in the gun. The valve shall be controlled by a trigger switch on the gun to provide instantaneous flow of shielding gas.

3.4.1.3 Liquid coolant valve. When the class B sets are to be used in conjunction with a standard water tap supply, the sets shall have a solenoid valve to control the flow of coolant through the gun and be designed to operate at pressures up to 80 pounds per square inch (PSI). The valve shall be controlled by the trigger switch on the gun, and may operate in conjunction with the gas valve specified in 3.4.1.2.

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3.4.1.4 Gun and cable overheat protection. The class B, liquid cooled, welding sets shall have a safety device to prevent the overheating of gun and cable assembly.

3.4.1.5 Inching switch. A switch shall be provided on the electric control unit or the gun to inch the wire while threading and adjusting the roll pressure on the wire. The switch shall be independent of the trigger switch. The gas supply and welding current shall not be energized while using the inching switch.

3.4.2 Wire feeder unit. The wire feeder unit shall consist of, but not be limited to, a wire drive, a feed roll pressure control, a wire feed drive motor, and a wire roll holder.

3.4.2.1 Wire drive. The wire drive shall have replaceable wire guides on the infeed and outfeed sides of the feed roll. The drive shall consist of one or more driven feed rolls, grooved, or knurled or serrated to grip and drive the wire. The wire feed rolls shall feed all classes and sizes of wire specified in table I with or without the necessity of shifting gears or by changing sizes of rolls/gears or motor gear assembly. The wire drive assembly shall be electrically insulated from the base and enclosure of the control unit. The wire feeder for the type II pull wire set may have provisions for driving and connecting a flexible shaft drive for driving feed rolls in a mechanically driven gun. The quantity and size of wire guides and gears to be furnished with the set shall be specified (see 6.2.1).

3.4.2.2 Wire feed roll pressure control. The set shall have a control to increase or decrease the pressure of the feed rolls on the wire. The control may have a quick release device to release the roll pressure and shall be resettable without affecting the pressure setting.

3.4.2.3 Wire feed drive motor. The motor shall be of adequate size to drive the wire feed at all speeds herein specified. The motor shall have sealed for life ball or roller bearings. The wire feed drive motor for type III, class A, style 2 shall be synchronized with an air driven turbine in the gun.

3.4.2.4 Wire feed braking. The wire feed shall stop upon the release of the trigger switch on the gun. Braking of wire feed shall be accomplished electrically or mechanically to quickly stop the wire feed to prevent overrun of the wire.

3.4.2.5 Wire holder. The sets shall have provision for mounting a disposable wire reel. The type I, class A, style 4 set shall have provisions for mounting an enclosed disposable wire reel with a capacity of not less than eight (8) pounds of steel wire. Installation of the wire reel shall be accomplished without the use of tools. When specified (see 6.2.1), the holder shall have the necessary brackets and adapters to accommodate large coils of wire. The adapters shall keep the coil concentric with the hub and provide a smooth payoff of wire. The holder shall have an adjustable device to put drag or

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tension on the hub so that the reel will stop turning when the wire feed stops. The hub shall be electrically insulated from the frame of the unit.

3.4.3 Gun.

3.4.3.1 Gun type I, push. The gun for the type I, class A air cooled sets shall consist of a hand-held gun with a straight, gooseneck or pistol type grip and control switch, and a cable assembly as described in 3.4.4. The gun and cable assembly shall be 10 to 15 feet in length. The guns shall be of the types and capacities to meet the requirements of table I. The gun shall be of lightweight, high-impact resistant construction, and shall be designed to be manually operated with one hand. All of the outside surfaces of the gun shall be electrically insulated from the welding current and the control circuit to prevent arcing if the gun is brought in contact with the work while welding. The internal passages for the shielding gas and coolant flow shall be designed to provide the proper flow without any leakage. The type I, class B liquid cooled gun and cable assembly shall consist of the same components as the type I, class A and, in addition, will have two (2) liquid coolant hoses. The type I, class B guns shall be compatible with permanent type anti-freeze solution.

3.4.3.1.1 Nozzle. The gun shall have a replaceable nozzle to direct the flow of shielding gas to completely shield the electrode and the work from the surrounding atmosphere. The quantity of nozzles to be provided with each gun shall be as specified (see 6.2.1). The nozzle or work end of the gun barrel shall be straight or curved as specified (see 6.2.1).

3.4.3.1.2 Contact tips and tubes. The gun shall have a replaceable contact tip or tube, straight or curved, to transfer the full amperage of the welding current to the electrode. The quantity and type of tips and tubes shall be as specified (see 6.2.1).

3.4.3.1.3 Control switch. The gun shall have a switch to start and stop the wire feed, the flow of shielding gas, the flow of coolant on the class B sets, and the welding current. The switch shall be designed to be manipulated by the thumb or forefinger of one hand and shall be the hold-on type.

3.4.3.2 Gun type II, class A, style 1, pull. The type II, class A, style 1 gun shall have provisions for the wire feeding into the rear of the gun. The gun shall be designed to be operated in all positions as a one-hand held unit and shall be capable of producing satisfactory weld when used in conjunction with either a constant current or constant voltage DC power source. The welding cables and the hose shall be not less than 50 feet long and the wire feed cables shall be not less than 10 feet long.

3.4.3.2.1 Drive motor and gear assembly and feed rolls. The motor shall be of the totally enclosed type, with sealed-for-life, ball or roller bearings. The feed rolls shall have means for adjusting pressure on the wire. The wire shall not be deformed by the feed rolls to the extent that it affects changes in the welding arc. The rolls shall be hardened for long service life.

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3.4.3.2.2 Wire spool housing. The housing shall be designed to hold a wire spool measuring 4 inches in diameter by 1-3/4 inches in width and an arbor hole of 5/8 inches diameter. The housing shall have means of fastening the spool in place and to prevent it from unwinding faster than the wire feed mechanism is feeding the wire. The housing shall have a removable or hinged cover to protect the wire from spatter, water, and dirt. The housing and cover shall be made of high impact resistant material.

3.4.3.2.3 Gun controls. A rheostat or potentiometer type wire feed speed control shall be furnished and shall be located either on the gun or on the electric control unit (see 3.4.1.1). One complete turn of the rheostat or potentiometer on the gun shall control the feed through its entire range. A grip type trigger switch shall be provided in the welding gun handle to start the motor of the electrode wire feed mechanism, the flow of inert gas, and welding current to the gun. A switch shall be provided on the gun for inching the wire without energizing the gas flow and welding current controls. The inching switch shall be located or guarded to prevent accidental closing.

3.4.3.2.4 Gas shielding nozzle and contact tube. The nozzle shall be designed to provide a continuous blanket of shielding gas to shield the weld zone from contamination by the surrounding atmosphere. When specified (see 6.2.1), curved nozzles and companion wire guides and contact tubes shall be furnished for welding in close or confined areas.

3.4.3.2.5 Cable and hose. The spool-on gun shall have not less than 30 feet of control cable, welding conductor cable, welding ground cable, and shielding gas hose. When required, longer cable and hose shall be provided in the length specified (see 6.2.1).

3.4.3.3 Gun, type II, class B, style 1 pull. The type II, class B, style 1 gun shall be the same as the type II, class A, style 1 gun except the guns shall have the wire feed rolls driven by a flexible shaft connected to the rear of the gun, and shall be water cooled. The gun shall have shaft, cable, and hoses not less than 10 feet long.

3.4.3.4 Gun, type III, push-pull. The type III gun shall conform to the requirements of paragraph 3.4.3.2. The gun shall be able to operate in conjunction with a wire feeder that has its wire feed synchronized with the wire feed in the gun. The gun shall have cable and hose not less than 30 feet long. Type III, class A, style 2 gun shall have an air driven turbine motor in the gun, synchronized with wire feeder.

3.4.4 Cable assembly. The cable assembly shall consist of the welding current conductor, the shielding gas hose, liquid coolant hoses and class B sets, wire guide conduit assembly, when applicable, and electric control cable. The welding current conductor, shielding gas hose, coolant hoses, and the electric control cable shall be bundled together and taped or clamped not less than every 18 inches, or shaped or molded to become one round integral cable, covered in a heat and abrasion resistant sheath.

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3.4.4.1 Welding current conductor. The welding current conductor shall be of the extra flexible type with a synthetic rubber or vinyl resin insulation. The insulation may fit loosely around the conductor to permit using it as a hose for the shielding gas flow to the gun or the coolant flow for the gun.

3.4.4.2 Shielding gas hose. The shielding gas hose in the cable assembly shall be made of high heat resistant vinyl resin or synthetic rubber. The hose shall be suitable for use with all of the gases used in the metal inert gas shielded and metal gas shielded welding process.

3.4.4.3 Liquid coolant hoses. The coolant hoses shall be made of high heat resistant vinyl resin of synthetic rubber. The hoses shall be compatible with permanent type anti-freeze solutions.

3.4.4.4 Wire guide conduit assembly. The wire guide conduit or casing shall be flexible to permit the feeding of the wires at all of the speeds herein specified and in all positions that it will be subjected to in welding operations. The inside surface of the flexible conduit or casing shall be designed to feed hard or soft bare wire with a minimum of friction. The conduit or casing may have a replaceable liner to adopt to the feeding of hard or soft bare wires. The liner shall be made of flexible, abrasion-resistant material.

3.4.4.5 Electric control cable. The control cable shall contain the conductors for the low-voltage control circuit. The connection on the end of the control cable for connecting to the electric control shall be polarized or keyed to prevent incorrect installation.

3.4.4.6 Type I, class A, style 4, cable and hose. The compact portable set shall have not less than 100 feet of the necessary cable and hose for connection of the welding set to the power supply and shielding gas supply. The 100 feet may consist of four 25 foot sections with connectors.

3.4.5 Interconnecting cords, welding cables, gas and coolant hoses. The welding sets shall have all necessary electric cords and cables, welding cables, shielding gas and coolant hoses for connection of the power supply, shielding gas supply, and coolant supply to the components of the welding set. Unless otherwise specified herein, the cables and hoses shall be not less than 25 feet in length. When separate shielding gas and coolant hoses are used, the inside diameter of the hoses shall be not less than 1/4 inch and the hoses shall have fittings with right-hand (rh) threads for the shielding gas connections, and left-hand (lh) threads for the coolant connections.

3.4.6 Enclosure. The welding sets shall have the electric controls completely enclosed. When the electric control, wire feeder, and the wire supply reel holder are mounted together on a common base, the electric control shall be in a separate shielded compartment or enclosure. The externally mounted reel holder shall be enclosed to protect the wire from dirt and contamination. The type II control enclosure shall be small enough to pass through a 13- by 18-inch rectangular opening. The type I, class A, style 4 set shall have the control and wire feeder totally enclosed in a weatherproof housing. The enclo-

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TABLE I - Equipment characteristics.

Type Class and Style	Wire Size Diameter in Inches		Amperage @ 70 Degree Ambient Temperature and Continuous Duty	Wire Speed Range Inches per Min.	
	Hard	Soft		Minimum Not More Than	Maximum Not Less Than
Type I					
Class A					
Style 1	.030-.045	.035-.046	200	60	600
Style 2	.045-.093	.046	400	60	500
Style 3	.078-.125	.046	400	80	400
Style 4	.0380-.045	.035-.046	200	60	500
Class B					
Style 1	.030-.045	.035-.046	250	60	600
Style 2	.045-.093	.093	500	60	400
Type II					
Class A					
Style 1	.030-.045	.030-.0625	200	110	400
Style 2	.030-.045	.030-.0625	200	120	600
Class B					
Style 1	.030-.0625	.030-.0625	400	110	400
Type II					
Class A					
Style 1	.030-.045	.030-.046	200	30	400
Style 2	.030-.035	.030-.045	200	30	400

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sure shall have a hinged or detachable cover to provide easy access to the wire supply reel and all other components.

3.4.7. Gas flowmeter and regulator. When specified (see 6.2.1), each welding set shall have a flowmeter and regulator to control the flow of shielding gas to the welding gun and to measure the cylinder pressure. The flowmeter shall have a scale calibrated for use with argon, helium, or CO_2 gas and shall have a flow rate of not less than 0-60 cubic feet (cfh) per hour. The flowmeter scale shall be protected by a clear high-impact resistant material, and shall be easily removed for cleaning or changing scales. The regulator shall have a gage to register the pressure in the gas cylinder and shall have gas connections for connecting to commercial gas cylinders.

3.4.8 Welding current contactor. When required (see 6.2.1), the welding set shall have a magnetically operated contactor to interrupt the welding current. The contactor shall be controlled by the trigger switch on the gun. The contactor may be mounted in a separate enclosure and positioned to remain vertical or horizontal when in operational use. The ampere rating of the contactor shall be as specified.

3.4.9 Portability. All the electric controls and wire feeders of the sets covered by this specification shall have a handle or other means of carrying to the work site. The components shall have provisions for hanging by a hook on a crane, hoist, boom, or overhead positioner. When required, the electric control, the wire feeder, and wire supply holder shall be mounted on a flat skid type base, a two (2) wheel dolly with a skid and handle, or a swivel mount for a flat surface on the top of the power supply as specified (see 6.2.1).

3.4.10 Self-contained liquid cooling system. When specified (see 6.2.1), the welding sets shall have portable, self-contained, liquid cooling, recirculating systems. The system shall consist of a tank, electric motor-driven pump, and a strainer or filter. The system shall be of adequate capacity to provide the proper cooling of the welding torch at all times. The self-contained cooling system shall be able to operate in ambient temperatures from 9° Fahrenheit (F) to $\pm$124° F.

3.4.11 Spot welding control. When specified (see 6.2.1), the type I and II welding sets shall be provided with a control and accessories to perform spot and plug welds. The timer and control shall be designed for use as a separate control or as integral with wire feed control. The control shall be provided with line and drive motor fuses. The guns shall have replaceable nozzles that will adapt to the spot and plug welding process. The number and types of nozzles shall be as specified (see 6.2.1).

3.4.12 Solid state components. Solid state design shall be used throughout for electronic components.

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3.5 Performance. The welding sets shall be capable of producing satisfactory welds on ferrous and nonferrous alloys within the capacities specified herein, when connected to a proper welding power source. The welding sets shall be capable of welding under any of the following processes.

a. Gas metal arc welding using argon, helium, or combination of argon and helium.

b. Gas metal arc welding using CO_2 or CO_2 and argon.

c. Tubular flux cored wire welding, with or without CO_2 for type I, class A, style 3.

3.6 Standard equipment. Unless otherwise specified, one set of hand tools, normally furnished by the manufacturer, shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.7 Marking of instruction plates, indicating plates, and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding set. The nameplate shall contain the information listed below. If the welding set is a special model, the model designation shall include the model of the basic model and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Electrical characteristics
Contract Number or Order Number
National Stock Number or Plant Equipment code
Date of manufacture

3.9 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagram, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.10 Workmanship. Workmanship of the welding sets and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall be a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection condition. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations or performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The welding sets shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

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4.7.1 Assembly line tests. The manufacturer's final assembly test of the equipment will be acceptable provided the tests and recorded performance data prove the equipment complies with the requirements specified herein.

4.7.2 Operational test. The welding set shall be operated without any wire feeding through the wire feeder for a minimum of 30 minutes, during which time the controls, the wire feed motor, and the gas flowmeter will be actuated and observed to determine if the set operates properly. After observing the action of flowmeter, the valve on the gas cylinder shall be closed. To keep the wire feed motor running for this test, the trigger on the gun may be tied down or a jumper wire placed on the proper terminals of the control circuit in the electric control.

4.7.3 Performance test. The following tests shall be performed on the welding sets.

4.7.3.1 Wire feed drive test. This test shall be applicable to all units with the exception of type II, class A, style 2. The wire shall be fed through the gun and cable assembly with all of the cable assembly coil to be 24-inch diameter ID. The direction or wind of coil shall be in the opposite direction of the wind of the electrode wire on the reel or wire holder. The wire feed drive shall be energized and there shall be no evidence of the wire feed motor being overloaded while feeding the wire through the conduit at the maximum rate shown in table I.

4.7.3.2 Welding set. The set shall be connected to a DC constant voltage power supply and a fillet-type weld 12 inches long shall be made using the following materials. Aluminum wire and plate type 4043 may be substituted wherever ER1100 aluminum is specified.

Type I, class A

Styles 1 and 2 - One weld using .045" mild steel wire and mild steel plates 3/8" thick. One weld using 3/64" ER 1100 aluminum plate 3/8" thick.

Style 3 - One weld using 5/64" diameter cored wire conforming to AWS A5.1-58T specifications and 1/2" mild steel plates.

Style 4 - One weld using .045" diameter mild steel wire and mild steel plate 3/8" thick.

Type I, class B

Style 1 - One weld using .045" diameter mild steel wire and mild steel plate 3/8" thick. One weld using 3/64" ER 1100 aluminum wire and 1100 aluminum plate 3/8" thick.

Style 2 - One weld using .045" diameter mild steel wire and

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Style 2 - One weld using .045" diameter mild steel wire and 3/8" mild steel plate. One weld using 3/32" ER 1100 aluminum wire and 1000 aluminum wire and 1100 aluminum plate 1/2" thick.

Type II, class A

Styles 1 and 2 - One weld using .045" diameter mild steel wire and 3/8" thick mild steel plate. One weld using 1/16" ER 1100 aluminum wire and 1100 aluminum plate 3/8" thick.

Type II, class B

Style 1 - One weld using 1/16" diameter mild steel wire and mild steel plate 3/8" thick. One weld using 1/16" ER 1100 aluminum wire and 1100 aluminum plate 3/8" thick.

Type III, class A

Style 1 - One weld using 3/64" diameter mild steel wire and mild steel plate 3/8" thick. One weld using 3/64" ER 1100 aluminum wire and 1100 aluminum plate 3/8" thick.

Style 2 - One weld using .035" diameter steel wire and one weld using .045" diameter ER 1100 aluminum wire on 3/8" mild steel and aluminum plates respectively.

4.7.4 Rough handling test. When specified (see 6.2.1), the following test shall be performed. Each set shall be dropped 10 times on a solid floor from a height of 6 inches. No particular points of impact are specified; but the dropping should simulate as near as possible the manner in which an outfit being handled with no particular care may be deposited on a deck in being moved from place to place. Afterward, there shall be no evidence of apparent damage. The welding performance tests specified in 4.6 shall be performed after this test is completed. The rough handling shall have resulted in no appreciable difference in performance.

4.8 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

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6. NOTES

6.1 Intended use. The welding sets covered by this specification, when connected to a DC power source of the proper type and size and a shielding gas/water supply, are to be used in performing semiautomatic arc welding of metals in all welding positions, in manufacturing, and repair and construction processing, both indoors and outdoors.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and style required, etc. (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), stated required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Compatibility with power supply, if required (see 3.4).

j. Quantity and sizes of wire guides and gears if required (see 3.4.2.1).

k. Wire holder, if different (see 3.4.2.5).

l. Specify quantity of nozzles required, and design (straight or curved) (see 3.4.3.1.1).

m. Quantity and type and tip/tube (see 3.4.3.1.2).

n. Curved nozzles and guide tubes, when specified (see 3.4.3.2.4).

o. Cables and hose length, if different specify total feet required (see 3.4.3.2.5).

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p. Gas flowmeter and regulator, when specified (see 3.4.7).

q. Welding contactor, size and type, when specified (see 3.4.8).

r. Portability, as specified (see 3.4.9).

s. Self-contained liquid cooling system, when required (see 3.4.10).

t. Spot welding control, accessories and nozzles, when required, specify (see 3.4.11).

u. Standard equipment, if different (see 3.6).

v. Nameplate, if different (see 3.8).

w. First article inspection, if different (see 4.4).

x. Quality conformance inspection, if different (see 4.5).

y. Rough handling test, if required (see 4.7.4).

z. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workpiece. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99
Navy - SH

Review activities:
Army - AL
Navy - OS, YD, SH
Air Force - 84
DLA - GS

User activities:
Army - CE, AV
Navy - MC, AS

Preparing activity:
DLA - IP

Project (3431-0170)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80102C	Welding Set, Arc, Gas Shielded, Consumable Electrode

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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